

The background is a blurred image of a computer monitor. It shows a dark-themed code editor with various lines of code in different colors (blue, green, white). At the bottom, a taskbar is visible with several application icons, including what looks like a web browser, a file explorer, and some development tools. The overall image is out of focus, serving as a backdrop for the text.

C#

Level 1

Two Dimensional Array Declaration



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Definition

Two-Dimensional Array

-  A 2D array stores elements in rows and columns.
-  It is commonly used for tables, matrices, and grids.

Explanation

Declaration

 `int[,]` declares a two-dimensional array.

 `new int[2, 3]` creates 2 rows and 3 columns.

💡 Example

```
int[,] numbers = { { 12, 13 }, { 55, 77 } };  
Console.WriteLine("2D Array Elements : ");  
for (int i = 0; i < numbers.GetLength(0); i++)  
{  
    for (int j = 0; j < numbers.GetLength(1); j++)  
    {  
        Console.Write(numbers[i, j] + " ");  
    }  
    Console.WriteLine();  
}
```

➡ Result

```
2D Array Elements :  
12 13  
55 77
```

Note

- ◆ For new `int[2, 3]`, valid rows are $0 \rightarrow 1$.
- ◆ Valid columns are $0 \rightarrow 2$.
- ◆ `[2, 0]` and `[0, 3]` are invalid indexes.
- ◆ Total elements = rows $\times$ columns.



Comparison

1D Array

- ✓ `int[]`
- ✓ Uses one index: `array[index]`
- ✓ Represents a list.

2D Array

- ✓ `int[,]`
- ✓ Uses two indexes: `array[row, column]`
- ✓ Represents a table or grid.

✦ Summary

- ✓ `int[,]` represents a 2D array.
- ✓ `new int[2, 3]` creates 2 rows and 3 columns.
- ✓ Indexes start from 0.
- ✓ Elements are accessed using `[row, column]`.
- ✓ Nested loops are commonly used to traverse all elements.

Thank you 😊